FILE: C:\HCL_TRAINING\server\src\services\jobIntelligence\jobAnalyzer.js

LINES: 201 | SIZE: 8.6 KB | LANGUAGE: JS

MODULE: 0-100% Job Readiness Score & Buzzword Detector

```
const VAGUE_PHRASES = [
  'rockstar',
  'ninja',
  'guru',
  'wizard',
  'wear many hats',
  'fast-paced environment',
  'work hard play hard',
  'self-starter who needs no direction',
  'competitive salary',
  'unlimited potential',
  'team player with good vibes',
  'jack of all trades',
];

/**
 * Analyzes a job posting for quality, clarity, and readiness
 */
function analyzeJobReadiness(jobData) {
  let score = 100;
  const missingElements = [];
  const detectedVaguePhrases = [];
  const suggestions = [];

  const textToScan = `${jobData.title || ''} ${jobData.rawDescription || ''} ${(jobData.
> responsibilities || []).join(' ')}`.toLowerCase();

  // 1. Mandatory Skills check
  if (!jobData.mandatorySkills || jobData.mandatorySkills.length === 0) {
    score -= 30;
    missingElements.push('No mandatory skills specified');
    suggestions.push('Add at least 3-5 mandatory skills to enable precise candidate matc
> hing.');
  } else if (jobData.mandatorySkills.length < 3) {
    score -= 10;
    missingElements.push('Fewer than 3 mandatory skills');
    suggestions.push('Consider defining more core requirements to distinguish strong can
> didates.');
  }

  // 2. Responsibilities check
  if (!jobData.responsibilities || jobData.responsibilities.length === 0) {
    score -= 25;
    missingElements.push('Missing key responsibilities');
    suggestions.push('Detail day-to-day deliverables and expectations.');
  }

  // 3. Experience check
  if (!jobData.minYearsExperience || jobData.minYearsExperience <= 0) {
    score -= 10;
    missingElements.push('Unspecified minimum experience duration');
    suggestions.push('Clarify minimum years of experience expected.');
  }

  // 4. Qualifications check
  if (!jobData.qualifications || jobData.qualifications.length === 0) {
    score -= 10;
    missingElements.push('No degree or educational qualifications listed');
    suggestions.push('Specify minimum education level or equivalent experience.');
  }

  // 5. Salary transparency check
  if (!jobData.salaryRange || (!jobData.salaryRange.min && !jobData.salaryRange.max)) {
    score -= 10;
    missingElements.push('Salary range not provided');
    suggestions.push('Providing a compensation band improves applicant quality and candi
```

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> date engagement.');
64 |   }
65 |
66 |   // 6. Vague wording detection
67 |   for (const phrase of VAGUE_PHRASES) {
68 |     if (textToScan.includes(phrase)) {
69 |       detectedVaguePhrases.push(phrase);
70 |       score -= 4;
71 |     }
72 |   }
73 |
74 |   if (detectedVaguePhrases.length > 0) {
75 |     suggestions.push(`Replace vague phrases (${detectedVaguePhrases.join(', ')}) with sp
> ecific, verifiable role criteria.`);
76 |   }
77 |
78 |   const clarityScore = Math.max(10, Math.min(100, score));
79 |   const isReady = clarityScore >= 70;
80 |
81 |   return {
82 |     clarityScore,
83 |     isReady,
84 |     missingElements,
85 |     vaguePhrases: detectedVaguePhrases,
86 |     suggestions,
87 |   };
88 | }
89 |
90 | /**
91 |  * Intelligent Job Enhancement / Generation Templates
92 |  */
93 | const DOMAIN_SUGGESTIONS = {
94 |   'full-stack': {
95 |     mandatorySkills: [
96 |       { name: 'React', minYears: 3, importance: 'critical', category: 'Frontend' },
97 |       { name: 'Node.js', minYears: 3, importance: 'critical', category: 'Backend' },
98 |       { name: 'TypeScript', minYears: 2, importance: 'high', category: 'Frontend' },
99 |       { name: 'PostgreSQL', minYears: 2, importance: 'high', category: 'Database & Stora
> ge' },
100 |     ],
101 |     preferredSkills: [
102 |       { name: 'Docker', importance: 'high', category: 'Cloud & DevOps' },
103 |       { name: 'AWS', importance: 'medium', category: 'Cloud & DevOps' },
104 |       { name: 'GraphQL', importance: 'nice-to-have', category: 'Backend' },
105 |       { name: 'Tailwind CSS', importance: 'medium', category: 'Frontend' },
106 |     ],
107 |     responsibilities: [
108 |       'Design, develop, and maintain performant full-stack web applications and microser
> vices.',
109 |       'Collaborate with product designers and backend engineers to implement intuitive u
> ser workflows.',
110 |       'Architect robust REST and GraphQL APIs with comprehensive automated test suites.'
> ,
111 |       'Ensure high code standards, security posture, and database optimization.',
112 |     ],
113 |     qualifications: ["Bachelor's Degree in Computer Science, Software Engineering, or eq
> uivalent practical experience"],
114 |   },
115 |   'devops': {
116 |     mandatorySkills: [
117 |       { name: 'Kubernetes', minYears: 3, importance: 'critical', category: 'Cloud & DevO
> ps' },
118 |       { name: 'Docker', minYears: 3, importance: 'critical', category: 'Cloud & DevOps'
> },
119 |       { name: 'AWS', minYears: 3, importance: 'critical', category: 'Cloud & DevOps' },
120 |       { name: 'Terraform', minYears: 2, importance: 'high', category: 'Cloud & DevOps' }
> ,
121 |     ],
122 |     preferredSkills: [
123 |       { name: 'CI/CD', importance: 'high', category: 'Cloud & DevOps' },
124 |       { name: 'Linux', importance: 'high', category: 'Cloud & DevOps' },
125 |       { name: 'Python', importance: 'medium', category: 'Backend' },
```

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126 |         { name: 'Monitoring', importance: 'medium', category: 'Cloud & DevOps' },
127 |       ],
128 |       responsibilities: [
129 |         'Architect and manage multi-region cloud infrastructure using Infrastructure as Co
    >   de (Terraform).',
130 |         'Optimize CI/CD release pipelines to achieve zero-downtime automated deployments.'
    >   ,
131 |         'Establish enterprise telemetry, alerting, and incident response runbooks.',
132 |         'Enforce security best practices across Kubernetes clusters and cloud VPCs.',
133 |       ],
134 |       qualifications: ["Bachelor's Degree in Computer Science, Information Technology, or
    >   equivalent experience"],
135 |     },
136 |     'design': {
137 |       mandatorySkills: [
138 |         { name: 'Figma', minYears: 3, importance: 'critical', category: 'Design & Product'
    >    },
139 |         { name: 'Design Systems', minYears: 3, importance: 'critical', category: 'Design &
    >    Product' },
140 |         { name: 'Wireframing', minYears: 2, importance: 'high', category: 'Design & Produc
    >   t' },
141 |         { name: 'User Research', minYears: 2, importance: 'high', category: 'Design & Prod
    >   uct' },
142 |       ],
143 |       preferredSkills: [
144 |         { name: 'Prototyping', importance: 'high', category: 'Design & Product' },
145 |         { name: 'Tailwind CSS', importance: 'medium', category: 'Frontend' },
146 |         { name: 'HTML5', importance: 'medium', category: 'Frontend' },
147 |       ],
148 |       responsibilities: [
149 |         'Lead end-to-end product design from generative user research to high-fidelity imp
    >   lementation specs.',
150 |         'Maintain and evolve the global multi-brand design system in Figma and code tokens
    >   .',
151 |         'Facilitate design reviews with executive stakeholders and cross-functional engine
    >   ering teams.',
152 |         'Conduct regular usability tests and synthesize user feedback into actionable prod
    >   uct iterations.',
153 |       ],
154 |       qualifications: ["Bachelor's Degree in Design, Human-Computer Interaction, or equiva
    >   lent portfolio evidence"],
155 |     },
156 |     'ai': {
157 |       mandatorySkills: [
158 |         { name: 'Python', minYears: 3, importance: 'critical', category: 'Backend' },
159 |         { name: 'PyTorch', minYears: 2, importance: 'critical', category: 'AI & Machine Le
    >   arning' },
160 |         { name: 'Large Language Models (LLMs)', minYears: 1, importance: 'high', category:
    >    'AI & Machine Learning' },
161 |         { name: 'NLP', minYears: 2, importance: 'high', category: 'AI & Machine Learning'
    >   },
162 |       ],
163 |       preferredSkills: [
164 |         { name: 'LangChain', importance: 'high', category: 'AI & Machine Learning' },
165 |         { name: 'Vector Databases', importance: 'high', category: 'AI & Machine Learning'
    >   },
166 |         { name: 'Docker', importance: 'medium', category: 'Cloud & DevOps' },
167 |         { name: 'FastAPI', importance: 'medium', category: 'Backend' },
168 |       ],
169 |       responsibilities: [
170 |         'Train, fine-tune, and evaluate deep learning and transformer-based language model
    >   s.',
171 |         'Implement enterprise RAG (Retrieval Augmented Generation) architectures with vect
    >   or indexes.',
172 |         'Build scalable inference APIs with low latency and streaming responses.',
173 |         'Ensure ethical AI practices, guardrails against hallucinations, and rigorous benc
    >   hmark testing.',
174 |       ],
175 |       qualifications: ["Master's Degree or PhD in Computer Science, Artificial Intelligenc
    >   e, or Mathematics"],
176 |     }
177 |   };
```

```
178 |
179 |  /**
180 |   * Generates tailored suggestions based on title or domain
181 |   */
182 |  function generateJobEnhancements(title) {
183 |    const lower = (title || '').toLowerCase();
184 |    let selected = DOMAIN_SUGGESTIONS['full-stack'];
185 |
186 |    if (lower.includes('devops') || lower.includes('cloud') || lower.includes('infra') ||
    >  lower.includes('sre')) {
187 |      selected = DOMAIN_SUGGESTIONS['devops'];
188 |    } else if (lower.includes('design') || lower.includes('product designer') || lower.inc
    >  ludes('ui') || lower.includes('ux')) {
189 |      selected = DOMAIN_SUGGESTIONS['design'];
190 |    } else if (lower.includes('ai') || lower.includes('ml') || lower.includes('machine lea
    >  rning') || lower.includes('data science') || lower.includes('nlp')) {
191 |      selected = DOMAIN_SUGGESTIONS['ai'];
192 |    }
193 |
194 |    return selected;
195 |  }
196 |
197 |  module.exports = {
198 |    analyzeJobReadiness,
199 |    generateJobEnhancements,
200 |  };
201 |
```